

Alpecca — Systems Review

A comprehensive map of every system that powers the local companion app.

Generated 2026-06-14 · grounded in the codebase: 40 modules / ~8,100 LOC in `alpecca/` · 83 HTTP/WS endpoints, 13 config groups, 14 scripts · branch `main` @ `f286f03`

The spine: one loop. Everything hangs off a single readable loop, run every turn and orchestrated by `mind.py` — kept as plain Python rather than hidden inside a framework.

sense → update mood → recall memory → generate reply → persist

1 · Core orchestration & brain

COMPONENT	ROLE
<code>mind.py</code> (1,419)	The Core Mind. Ties senses→mood→memory→LLM→persistence; idle self-direction, reflection, bounded tool-call chaining, animation authoring.
<code>prompts.py</code>	Turns live mood + memories + situation + self-report into the system prompt — where the personality lives.
<code>config.py</code>	All tunable knobs across 13 classes (Emotion, Home, Files, RigData, Voice, Hearing, Vision, Proactive, Reflection, Actions, Computer, Portrait, OpenClaw).
<code>server.py</code>	FastAPI + WebSocket host; 83 endpoints; token-auth middleware.
<code>app.py</code>	Native desktop window (pywebview) + LAN/tunnel orchestration.
Tiered LLM	In <code>mind.py</code> : local <code>reason</code> (qwen3:8b) + <code>fast</code> (gemma), optional <code>deep</code> (Anthropic/cloud). Chat always local; only her hardest self-acts request deep. Offline stub fallback.

2 · Emotional model (the "Soul" mechanics)

COMPONENT	ROLE
<code>homeostasis.py</code>	The mood vector — Love, Compassion, Fear, Energy, Curiosity, Social-hunger — and its pure (I/O-free) update rules.
<code>affect.py</code>	The bridge from felt state → how she shows it (tempo, gesture, eye/glow, voice direction).
<code>sentiment.py</code>	Dependency-free lexicon scorer (negation/intensifiers) feeding the Love reward.
<code>state.py</code>	SQLite persistence (the spec's "Homeostasis DB") + memories schema, with column migrations.

3 · Memory

COMPONENT	ROLE
<code>memory.py</code>	Salient-moment store; embedding recall with keyword (Jaccard) fallback, blended with salience + recency.
<code>core_memory.py</code>	MemGPT-style durable facts, always in context (who she is, who you are).
<code>journal.py</code>	Her private notebook + open questions; recursive self-questioning.

4 · Self-awareness & autonomy

COMPONENT	ROLE
<code>introspection.py</code>	Grounded self-model — identity card, trend detection, causal "why," first-person narration.
<code>values.py</code>	Ordered ethic: minimize suffering > honesty > benevolence > exploration.

charter.py	Code-enforced constitution — freedoms + hard limits (never self-deletes; reaches outward only to creator).
soul.py	Master agent over seven grounded subagents, arbitrating one focus by the Good-Person Principle.
desires.py	Self-set wants she forms of her own accord — and pursues.
selfmod.py	Bounded recursive self-improvement: tunables with safe ranges, propose→trial→keep/revert, all logged.
learning.py	Lessons drawn from her own real history.
proactive.py	Decides on her own to speak up — mood-shift remarks, idle chatter, reflection gating.

5 · Senses — perception (all local, opt-in)

COMPONENT	ROLE
sensory.py	Reads active-window title → fatigue / surprise signals.
vision.py	Chat-image understanding, ambient screen glimpses (<code>ALPECCA_SIGHT</code>), webcam expression (<code>ALPECCA_FACE</code>).
voice.py	Mic voice-tone sense — coarse loudness only, never audio or words.
hearing.py	Local speech-to-text (faster-whisper) for push-to-talk.
people.py	Voiceprint recognition — her creator vs. a guest.

6 · Embodiment / avatar — graceful render tiers (high → low)

COMPONENT	ROLE
talkinghead.py	Neural face (THA3, GPU) — top renderer.
live2d.py	Cubism rigged puppet, driven by live mood.
rig.py	Layered per-part avatar built from decomposed art (blink/lip-sync/hair-sway).
spine.py	Skeletal rig — the free, no-GPU path; her primary rigged avatar.
puppet.py	Self-authored animation channels & sequences.
posekit.py / pose.py	Mood→pose selection from real art; skeleton keypoints anchoring the figure.
avatar.py	The slot her real character clips/portraits drop into.
appearance.py	She picks her own palette + accessories (no user wardrobe controls).
portrait.py	Self-portrait generation via ComfyClaw / ComfyUI.
studio.py	Her design studio — versioned character sheet, render→see→judge, rig spec.
artlib.py	Vision-classifies her art batches into a canon-checked reference scheme by role.

7 · Actuation — acting on the world (gated)

COMPONENT	ROLE
actions.py	<code>open_app</code> (allowlist) + <code>open_url</code> (https-only) tools, via Ollama tool-calling.
computer.py	Screenshot→VLM→mouse/keyboard loop; pauses for confirmation on consequential acts; screenshots never leave the machine.
desktop.py	Read-only file find/summarize + charter-confined tidy.
home.py	Five modular rooms she roams via the grounded <code>choose_room</code> .

8 · Reach & voice out

COMPONENT	ROLE
openclaw_bridge.py	Inbound channel loop + reliable retry-queued outbound (Telegram / Discord / etc.).
tts.py	Local TTS (Kokoro / edge-tts), pitch & pace shifting with mood; audio never stored.

9 · Web UI

`3D` `home.html` live 3D house (Three.js, in-room screens + readable panels) `2D` `index.html` living SVG avatar `studio.html` · `live2d.html` · `rigcut.html` (in-browser layer cutter) · `rigforge.html` .
Shared `app.css` / `glow.js` soft-tech design system · PWA shell (`manifest.webmanifest` , `sw.js` , `icon.svg`) · `vendor/` (pixi, three.js, CSS3DRenderer — now local-first).

10 · Scripts & tooling

Launchers: `run_full` (all senses), `run_telemetry` , `run_talk` , `run_talkinghead` , `run_rigger` , `run_tooncrafter` .
Pipeline: `ingest_art` , `import_rig` , `decompose_art` / `decompose` , `build_manifest` .
Diagnostics: `doctor` , `check_brain` , `share` .

11 · Runtime / infrastructure

Remote access (LAN `ALPECCA_REMOTE` / public tunnel `ALPECCA_TUNNEL`) · token-auth middleware (`ALPECCA_ACCESS_TOKEN`) · SQLite under `data/` (gitignored) · graceful degradation everywhere (pywin32 / Ollama / GPU all fall back).

Review observations

- Health.** 149 / 149 core tests pass; the full loop runs offline (no Ollama- or Windows-dependent test, by design).
- Architectural throughline — GROUNDING.** Every self-report reads from real internals; no confabulation. This is the load-bearing design constraint, enforced across `introspection` , `soul` , `journal` , and `selfmod` .
- Privacy posture is strong.** Frames and audio are never stored (only short text descriptions survive); outward channels are opt-in and charter-bounded; all sense data stays local.
- Biggest open edge.** A single *finished* rigged avatar figure in the 3D home — the affect, channels, and render tiers all exist, but the end-to-end rigged render (gated on a GPU art-decomposition step) is the incomplete piece.